EMMANUEL KIMUTAI CHEBII

Telecommunications & Information Engineering Undergraduate

chebiiemmanuel86@gmail.com | +254 708 444 950 | Eldoret, Kenya

PROFESSIONAL SUMMARY

Fourth-year Telecommunications and Information Engineering undergraduate at Dedan Kimathi University of Technology with practical experience in telecommunications, FTTH networks, fiber-optic installation, network configuration, and troubleshooting. Skilled in IoT systems, embedded technologies, and real-time monitoring, with additional knowledge of Python, machine learning, and computer vision. Completed Artificial Intelligence training through ICT Authority Kenya in collaboration with Pathways Technologies in July 2026. Eager to gain further hands-on experience, participate in engineering projects, and learn from experienced professionals in the telecommunications and IT infrastructure industry.

EDUCATION

Bachelor of Science in Telecommunications and Information Engineering

Dedan Kimathi University of Technology, Nyeri, Kenya

Undergraduate — fourth year

Kenya Certificate of Secondary Education (KCSE)

Chewoyet High School, Kapenguria | December 2022 | Grade: B+

TECHNICAL SKILLS

1. **Telecommunications and Network Infrastructure:** Telecommunications systems, network infrastructure, wireless networking, and connectivity solutions.
2. **Fiber-Optic Communication:** Fiber-optic communication systems, FTTH installation, fiber-optic cable handling, basic splicing, and ONU/OLT fundamentals.
3. **Network Configuration and Troubleshooting:** Network setup, configuration, fault diagnosis, and troubleshooting.
4. **Networking Technologies:** MikroTik router configuration, PPPoE authentication, VLAN fundamentals, IP addressing, and Wi-Fi hotspot configuration.
5. **Servers and Data Centers:** Basic understanding of server infrastructure, data center operations, and network connectivity.
6. **Wireless Communication:** Wireless networking technologies, Wi-Fi configuration, and connectivity troubleshooting.
7. **Python and Machine Learning:** Python programming fundamentals, machine learning concepts, and practical AI project development.
8. **Computer Vision and AI:** Object detection, optical character recognition (OCR), object tracking, and computer vision applications.
9. **Internet of Things (IoT) and Embedded Systems:** ESP32, Arduino, sensor integration, embedded-system development, and real-time monitoring.
10. **Database and Real-Time Applications:** Firebase Realtime Database fundamentals, data storage, and real-time monitoring applications.
11. **Hardware and Electrical Systems:** PCB fabrication and design exposure, RJ45 crimping, T568A/T568B wiring standards, and basic electrical installation.

INDUSTRIAL ATTACHMENT

Kimberley Telecommunications Network — Kapsabet, Kenya

Industrial Attachment | 2 March – 15 May 2026

- Assisted with FTTH drop-cable installation, cable routing, fiber handling, and customer-premises connectivity.
- Gained practical exposure to fiber-optic splicing and installation safety procedures.
- Configured home routers and Wi-Fi hotspots, including SSIDs, passwords, and basic network parameters.
- Assisted with troubleshooting ONU/router connectivity issues and monitoring network performance.

- Learned about MikroTik PPPoE client configuration, user authentication, VLANs, and bandwidth management.

ACADEMIC & PRACTICAL PROJECTS

A).Flood Monitoring System (IoT)

Developed/planned an ESP32-based water-level monitoring system using an ultrasonic sensor, with Firebase integration and SMS alert functionality as project objectives.

B).Arduino-Based Home Automation & Laboratory Work

Academic exposure to Arduino experiments, sensor-based automation, PCB fabrication, and basic electrical installation.

C).License Plate-Based Vehicle Tracking System

Python | YOLOv8n | EasyOCR | ByteTrack | Computer Vision Developed a computer-vision workflow to detect vehicles and recognize license plates in recorded video, combining YOLOv8n object detection, EasyOCR character recognition, and ByteTrack multi-object tracking. Designed the system to produce searchable vehicle records with plate numbers, timestamps, and entry/exit information.

TRAINING & PROFESSIONAL DEVELOPMENT

Artificial Intelligence Training Programme

ICT Authority Kenya in collaboration with Pathways Technologies | Completed July 2026

PROFESSIONAL STRENGTHS

1. **Problem-Solving and Analytical Thinking:** Ability to analyze technical challenges, identify potential solutions, and approach problems logically.
2. **Teamwork and Communication:** Effective communication and collaboration with team members to achieve shared objectives.
3. **Adaptability and Flexibility:** Ability to adapt to new technologies, changing environments, and different technical requirements.
4. **Technical Troubleshooting:** Ability to identify, investigate, and resolve basic networking, hardware, and system-related issues.
5. **Continuous Learning and Development:** Willingness to learn new technologies, improve technical competencies, and develop professionally.
6. **Practical and Hands-On Skills:** Ability to participate in practical engineering tasks, equipment setup, system configuration, and technical implementation.
7. **Technical Project Collaboration:** Experience collaborating on academic and technical projects involving telecommunications, networking, artificial intelligence, and IoT systems.

REFERENCES

Eng. Ronald Kosgei
Phone: +254 701 663 472
Address: Box 878, Nakuru
Email: ronkosgei@gmail.com

Mr. Beadan Njoroge
Phone: +254 711 257 650
Address: Private Bag, Kapenguria
Email: bmureithi86@gmail.com

Diane Chepchumba
Phone: +254 727 236 097
Address: P.O. Box 4157, Eldoret
Email: chebiidian@gmail.com